

Designing WindCage- Unpacking the Thinking and Prototyping a Propeller-Based Haptic Unit

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ABSTRACT

The topic of Augmented Reality Enabling Superhuman Sports (ARES) highlights the necessity of applying tangible game objects to achieve rich haptic feedback and realistic interactions. In this paper, we propose WindCage, a tetrakaidecahedron shape propeller-based haptic unit, and its two usages to provide different ways to experience "holding the wind." Users are required to execute physical movements and activities during the Superhuman Sports and Serious Game (SSSG) content. However, it is foreseeable that users' physical and cognitive condition varies. Therefore, we iteratively developed the system according to the Inclusive Design methodology to enable the maximum number of users with different abilities to adopt it in various scenarios. We aim to utilize the results to contribute insights into designing game objects in the Superhuman Sports and Serious Game community.

Keywords: Augmented virtual reality, inclusive design, tangible game object, haptic, force feedback.

Index Terms: Human-centered computing-Accessibility-Accessibility theory, concepts and paradigms

1 INTRODUCTION

The vision of Superhuman Sports is to reinvent sports that anyone can enjoy without the limitation of time and location [1,2]. Thus, technology is undoubtedly adopted to achieve its goal—precisely, those technologies that can bridge reality and virtual environments such as tracking, communication, and simulations. Nowadays, we can enjoy many sports anytime and anywhere through advanced technologies. For instance, Just Dance Now [3] allows users worldwide to compete with each other; on the other hand, due to the Covid-19 pandemic, even the most exciting Formula One World Championship is launched online as Formula 1® Esports Series [4]. It is evident that the users range from professional players to ordinary people and even physically challenged people. Therefore, besides breaking through the boundaries of time and location, here, we want to explore the other mission of the

Superhuman Sports genre, which is "for anyone can enjoy." The topic of overcoming somatic restrictions of the human body is deeply studied in the design and ergonomics field.

The general concept of "design for all" is continuously discussed in the conventional product design community. There have been several design approaches exist for covering the majority of potential users, especially for those whom with age-caused and congenital impairments, such as Accessible Design [5], Barrier-free design [6], Universal Design [7], and Inclusive Design [8]. These design theories share the same goal to design for everyone but start from a different point. According to the definition, accessible design takes disabilities as the priority; barrier-free design emphasizes eliminating the barriers of surroundings; universal design yearn for the greatest extent possible of products and environments for all people. Furthermore, inclusive design serves as a design methodology that enables and reaches the maximum range of human diversity.

Concurrently, propeller-based interaction devices have flourished as force-feedback haptic devices [10-15] and tangible game objects [16, 17] within Superhuman Sports and the somesthetic design field [19, 20]. Moreover, with the advanced tracking technology and the sensitive performance of DC brushless motor, the reaction of equipped propellers devices can almost be in real-time response to the user's movements. We see an opportunity to investigate how we can design propeller-based interaction devices that serve as tangible game objects that everyone can use in both Superhuman Sports and serious games (SSSG).

To explore this opportunity, we present WindCage (see Figure 1): a tetrakaidecahedron shape propeller-based haptic unit which can provide different ways to experience "holding the wind". WindCage builds on the related HCI works and is grounded in the inclusive design methodology. It can instantiate how we can weave design theory, conventional product engineering, and augmented virtual reality technology into a propeller-based interactive system.

Our main contributions include: (1) Instantiate the inclusive design with WindCage and deploy it in the SSSG field. (2) Showcase a VR instance of WindCage. (3) Design implications aim to inform close-range propeller-based interaction device designs.

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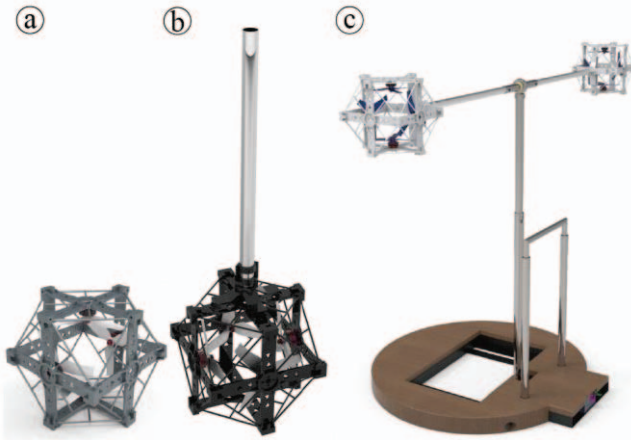


Figure 1: WindCage and its two usages.

2 BACKGROUND AND RELATED WORKS

2.1 Wind in Virtual Reality

Air is full of every corner in the real world. We could take air as the biggest, invisible environment surrounding us. Furthermore, the wind is the outcome of air flowing. Hence, it is natural that humankind is sensitive to the wind because it can offer us slight clues of changes around us. We have several ways to distinguish where the wind is, such as: (1) Vision: to see the deformation of movement on the surface of objects; (2) Hearing: to hear the noise of wind; (3) Cutaneous: to feel the flow of wind; (4) Kinesthesia and proprioception: to balance our body in the strong wind with the movement that you can behave. As wind can activate many perceptions, it can inform us from four channels. HCI researchers utilize wind's four primary modalities to create a realistic experience in virtual reality innovations. Wind-blaster [10] is worn on users' wrists and provides force feedback while shooting a gun in the VR game. Thor's hammer [11] installed propellers in a hammer-like device to generate propulsive force simulating feedback of water flow, sheep tugging, and pushing buttons. LevioPole [12] provides full-body force feedback by connecting eight propellers with a 1-meter aluminum pole. Its large scale offers the feeling of doing extreme exercises, such as kayaking [16] and super-leaping [15].

2.2 Toward Serious Game

The game objects in the serious game nowadays are primarily designed as controllers. We can notice the lack of mutual feedback within the serious game from the previous research cases. This could be owed to the design of game objects, mostly providing static physical properties such as shape, weight, etc. Thus, from the aspects of interaction, we can say the ways of communicating with the game are quite limited. Vibration is the most common haptic feedback of the commercialized product, such as Nintendo® Switch [13].

However, some research reports the chances to enrich the users' perception of the serious game genre. Aero-plane [14] is a handheld size force-feedback device which can render weight motion illusion. Virtual Kayaking experience [16] can also utilize a full-body haptic device to attract users to exercise more. Drone Chi [19] explore the opportunity to execute Tai Chi movements via drone as a medium.

3 CRYSTALIZED IN THE WINDCAGE

Our previous research results indicate that propeller-based haptic device has great potential to apply to SSSG field [12, 14, 15]. In the beginning, we start with technology to create a novel experience. After opening to any user for collecting feedback, we think that it is necessary to review from a design perspective to achieve the goal of "for anyone can enjoy". We aim to reshape the technology of propeller-based haptic devices as a universal tangible game object in Superhuman Sports and serious games. Hence, we adopt the Inclusive Design Cube (IDC) methodology [8] to weave the tangible user interface and various considerations into our proposed WindCage. We construct a cube of synchronization (see Figure 2) for WindCage to cover the maximum number of users in different scenarios. According to the five levels of IDC, the first and the fifth levels are user needs and usability testing. We would like to employ level 2-4 to organize our consideration into 3 axes: Level 2- user perception, Level 3- user cognition, and Level 4- user motor function.

As for each ax detailed hierarchy, we borrow the concept of fidelity from the AIP cube [9]. Fidelity plays a crucial role in the sense of presence. We want to adopt the fidelity to connect the Level 2,3,4 axes. The relationship of perceptions on Level 2 could be noted as follows: vision can offer the majority of fidelity in the system. We take vision as the light in the real world. Once the system provides visual feedback, the users can quickly map their personal experience to our designed system. However, what if we would like to enhance the interaction to a sophisticated level to match the user motor ax. Then, the passive and active haptic feedback should be considered. We synthesis IDC and AIP cube into a cube of synchronization to have a comprehensive viewpoint when designing the WindCage. This cube can help researchers consider the number of target users, their condition of physical ability, and the system capability simultaneously.

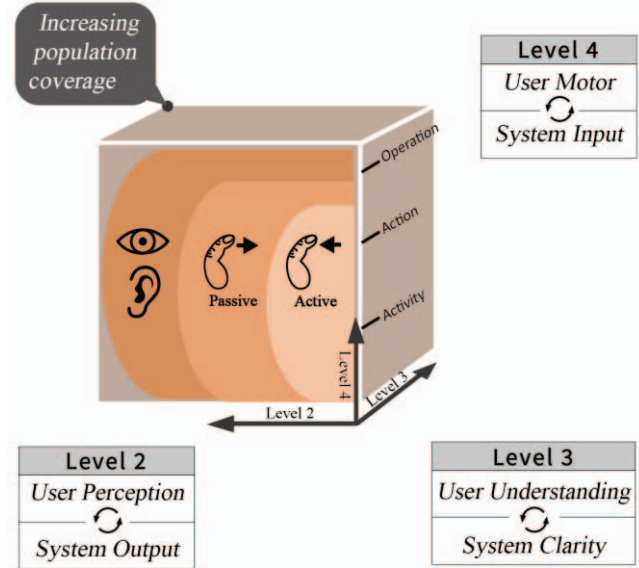


Figure 2: Cube of synchronization.

3.1 "Holding the Wind" through WindCage

Based on the IDC methodology, the optimized design needs critical feedback to achieve. Our goal is properly importing the propeller-based haptic unit to SSSG. Thus, we begin by investigating the more rigorous field- serious game in medical

scenarios. To comprehensively understand the scenarios of therapy, a focus group (see Figure 3) has been held to acquire insights from medical experts. The interviewed experts range from physicians to occupational therapists.

At first, we discussed the role of the tangible object in the serious game and its feasibilities. The rehabilitation training nowadays usually utilizes simple props to assist the therapies, such as a cup, a pen, or a sandbag. There is a fundamental principle in the practical rehabilitation process- most of the activities will be told as a task. For instance, when the therapist would like to ask the patient to stretch to his/her left side, they will ask them to reach the light of laser pointer or some tangible objects rather than directly command. The reason for assigning tasks in the therapy is that the goal of therapy is helping the patient execute their daily living activities. Moreover, in our daily activities, there exist no people to tell you how to finish a task step by step. It requires our actively figuring out the strategy. Furthermore, through such a process, the therapists can observe the progress of the patient at the same time and make a personalized task for the next stage.

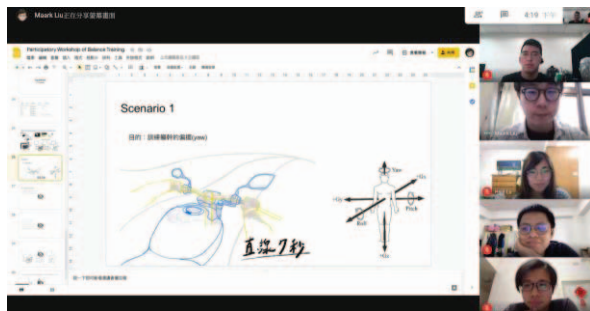


Figure 3: Focus group for mining requirements

We have discussed the relationship between the wind's characteristics and different human perceptions in the serious game. Finally, we consider what the ideal experiences of SSSG are. Both of them require numerous kinesthesia and proprioception. Thus, "holding the wind" through WindCage is determined as the core experience to effectively activate users' movements by sensing a virtual environment's force.

3.2 Product Design Engineering

Product design engineering is highly considered to make sure this game object could implement in the real field for further study. Therefore, efficiently manufacturing, assembling, and maintaining WindCage is the most crucial issue.

3.2.1 Refinement of Tetrakaidecahedron Shape

The propeller unit is the most critical and easily broken part. Hence, we design the propeller unit as the same module, and the tetrakaidecahedron shape is designed to minimize the size of WindCage. Furthermore, the extensibility of WindCage is considered. The structure of the module plate has been designed for attaching trackers and other sensing modules to augment various data from reality to the game. Benefit from the tetrakaidecahedron shape. There are up to fourteen surfaces of WindCage for us to extend the function (see Figure 4).

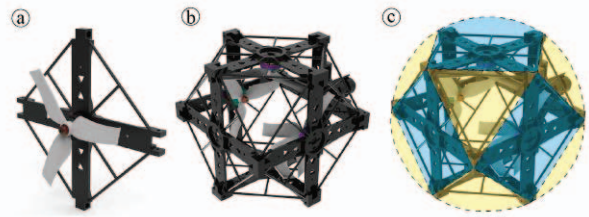


Figure 4: The propeller unit and WindCage.

3.2.2 The Integration of WindCage Supported Experience

We illustrate the architecture of WindCage supported experience (see Figure 5). The block of the haptic device is divided into WindCage and product design engineering. As a source of haptic feedback, we recognize that WindCage can theoretically generate force from any angle anywhere. Therefore, the follow-up issue is how we design the proper tangible object to integrate with WindCage to become a new experience in the SSSG genre.

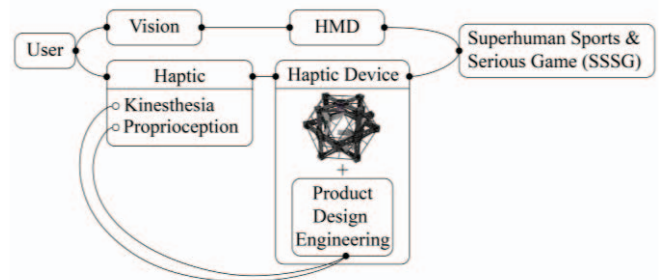


Figure 5: The integration of WindCage supported experience.

As for the power of WindCage, the adoption of a power supply or LiPo battery has been considered since the brushless motors generate the instantaneous high current. However, in a practical scenario, the general capacity of the LiPo battery is not suitable for over an hour of run times. From the usability perspective, frequently changing batteries is unreasonable in the practical situation and will break the user experience. Therefore, we employ a power supply for the WindCage system (see Figure 6). The specification of the DC brushless motor is 2300KV, 3-4S, 500W(max). The maximum force per motor can generate 1.8N force feedback within 0.35 seconds and requires 13.25A instantaneous high current. Thus, we adopt the power supply with 1000w, 12.15V, 82.3A specifications in the system.

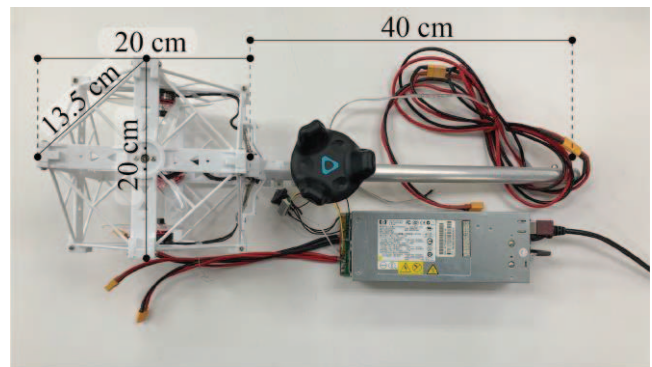


Figure 6: Power supply and the measurement of WindCage.

Based on the LevioPole [12] project and its supported experiences, we found a critical problem in designing a universal propeller-based force feedback tangible object for SSSG: the

device's weight could be a challenge to those whose physical condition is lower than average. As we can see in Fig 7, the single-hand version weighs 650g; the paddle-like version is around 1,800g. Therefore, consider that some users in the serious game might be tired before they finish the training. We design an arcade game machine-like version for this situation. The training-needed users can execute shoulder exercises with very mild loads to their bodies. The arcade game machine version could be applied to the paragliding and kayaking scenarios as a controller with force feedback in SSSG field. The direction of force feedback in these three versions is shown in Figure 7. In the next chapter, we will introduce a use case to inform the implications of the close-range propeller-based interaction device designs.

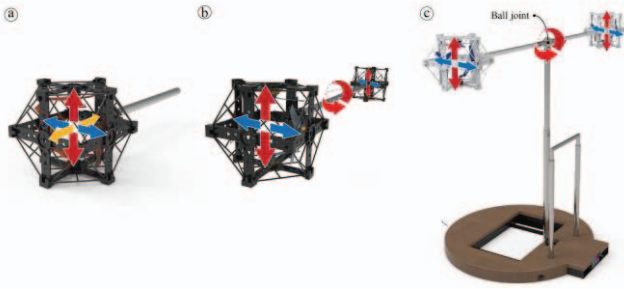


Figure 7: Force feedback pattern in the different setup

4 IMPLEMENTATION: A USE CASE

In order to better explore the possibility of applying WindCage to the SSSG, we determine to build a full-body activity with the competitive characteristic. Therefore, we present a fish scoop game powered by WindCage as a use case.

4.1.1 The Integration of WindCage Supported Experience

The fish scoop experience is built for training the balance ability of the user. Thus, the users are asked to scoop fish from one water tank to the other via rotating their trunks without moving their feet. In this case, we start by anchoring the point on the Level 4 ax (user motor) to adjust this cube to increase population coverage. Level 2 (system output) is already defined since the WindCage can provide perception, including vision, hearing, active and passive haptic feedback. Therefore, we focus on Level 3 (system clarity) to improve the level of match between the system and the real world. Then, after brainstorming about the interaction context, the fish scoop scenario is proposed to enhance the system clarity. The interaction scenario could let users comprehend how to use the game object and construct their moving strategy naturally.

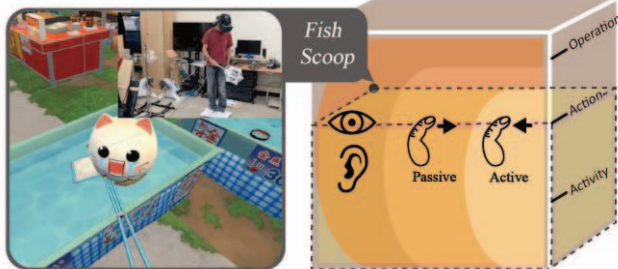


Figure 8: Fish scoop and its location in the cube of synchronization.

4.1.2 Further Development towards Practical Therapy

The core ability of balance training is to shift our center of mass (CoM) smoothly in the 3D space. As for the level of difficulties,

we are inspired by the multidimensional difficulty setting of a game concept for Alzheimer's patients [21]. The condition of users varies. Some users might have difficulty shifting their CoM from vertically, while others may face problems on the left to right. Therefore, we have set the difficulty parameters parallel rather than inheriting the difficulties higher and higher so that we can achieve the goals covering more users and motivating them by proper level of difficulty.



Figure 9: Unity as the engine to complete the fish scoop game.

5 CONCLUSION

Through this paper, we present WindCage and our iterative design process, which synthesize propeller-based haptic unit, conventional product design engineering, and inclusive design methodology [8]. We unpack our thinking of integrating our consideration into a cube of synchronization. We aim to utilize the results to contribute insights into designing game objects in the Superhuman Sports and Serious Game community.

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